

Test Yourself

Microeconomics

Income Elasticity of

Demand

Section A

Duration of each question is **40 mins**

1. Explain, with the help of examples, the significance of the size and sign of the coefficient of income elasticity of demand for the classification of a good **and** consider how confident you are of this classification. (8) **q2a/23/M/J/23**
2. With the help of a formula, explain the meaning of income elasticity of demand **and** consider the extent to which a rise in income will increase the consumption of **all** goods and services. (8) **q3a/22/F/M/24**
3. With the help of a formula, explain what is meant by the income elasticity of demand for a product **and** consider the extent to which demand for the product will always rise at the same rate as the income of its consumers. (8) **q3a/21/M/J/24**

Section B

Duration 20 mins

1. The table shows how an individual's weekly consumption of biscuits and coffee varies with income.

income (\$)	biscuits (packs)	coffee (cups)
100	0	5
150	5	10

Which statement about the income elasticity of demand over the range of income shown is true?

- A For biscuits it is greater than 1.
- B For biscuits it is zero.
- C For coffee it is less than 1.
- D For coffee it is unitary.

2. Public transport in an economy has an income elasticity of demand of -0.36 .

What does this mean about public transport?

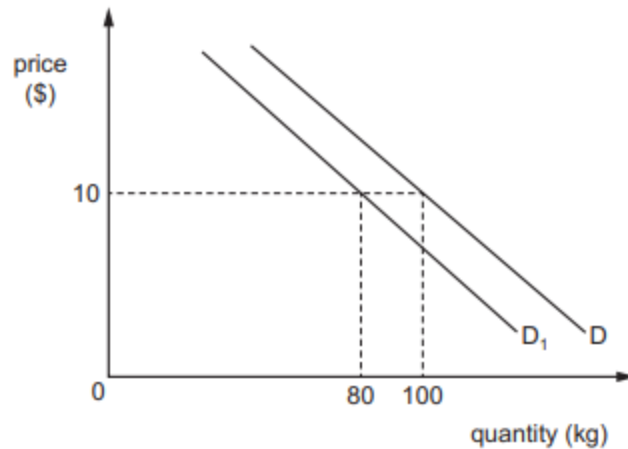
- A It has close substitutes.
- B It is a necessity.
- C It is a normal good.
- D It is an inferior good.

3. After a 10% decrease in a consumer's income, the consumer's demand for a good rises from 100 to 105.

What is the value of the consumer's income elasticity of demand for this good?

- A -2.0 B -0.5 C $+0.5$ D $+2.0$

4. The diagram shows how a 10% increase in consumer incomes shifts the demand for good X from D to D_1 .



Which statement about good X is correct?

- A** It is a normal good with income elastic demand.
 - B** It is a normal good with income inelastic demand.
 - C** It is an inferior good with income elastic demand.
 - D** It is an inferior good with income inelastic demand.
5. The table shows changes in a consumer's expenditure on various goods when her income increases from \$20 000 to \$24 000.

good	income \$20 000: amount spent on good (\$)	income \$24 000: amount spent on good (\$)
W	100	96
X	100	100
Y	200	224
Z	200	248

Assuming all else remains unchanged, for which goods is the consumer's income elasticity of demand greater than 1.0?

- A** W, Y and Z
- B** W and Z only
- C** W only
- D** Z only

6. When the incomes of consumers of good X fall by 5%, under which conditions will the demand curve shift furthest to the right?

	nature of good X	income elasticity of demand for good X
A	inferior good	-0.8
B	inferior good	-1.2
C	normal good	+0.9
D	normal good	+1.5

7. Which statement is correct?

- A** Demand for an inferior good has a positive relationship to income and a negative relationship to price.
- B** Demand for an inferior good has a negative relationship to income and a negative relationship to price.
- C** Demand for a normal good has a positive relationship to income and a positive relationship to price.
- D** Demand for a normal good has a negative relationship to income and a positive relationship to price.

8. A firm calculates that the income elasticity of demand for its product is -3.0.

What can be deduced from this information?

- A** In a period of economic growth, its total revenue should increase.
- B** In a period of economic recession, its total revenue should increase.
- C** It has few substitutes so it should increase its price to increase total revenue.
- D** It has many substitutes so it should decrease price to increase total revenue.

9. What is most likely to cause the demand curve of an inferior good to shift to the right?

- A** a rise in consumers' incomes
- B** a rise in income tax
- C** a rise in sales tax
- D** a rise in the price of a complement

10. What follows if the income elasticity of demand for a good has a value of -0.2 ?
- A When income rises less of the good is bought.
 - B When income rises more of the good is bought.
 - C When price falls more of the good is bought.
 - D When price rises less of the good is bought.

Question	Answer	Marks
1	Explain, with the help of examples, the significance of the size and sign of the coefficient of income elasticity of demand for the classification of a good and consider how confident you are of this classification. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) Knowledge of what is meant by income elasticity of demand (1) Formula for calculation of income elasticity of demand (1) Meaning of elastic/inelastic income elasticity of demand (1) AO2 Analysis (max 3 marks) Analysis of the significance of the size and sign of the coefficient for normal goods with relevant examples (1) Analysis of the significance of the size and sign of the coefficient for necessity goods with relevant examples (1) Analysis of the significance of the size and sign of the coefficient for inferior goods with relevant examples (1) Analysis of the significance of the size and sign of the coefficient for superior goods with relevant examples (1) Award maximum of 1 mark for two types of good but with no examples. Award maximum of 2 marks for three types of good but with no examples. AO3 Evaluation (max 2 marks) For a brief comment such as • Classification is dependent on income level of individual/household in case of inferior goods or necessities or superior goods • A necessity for a low-income household could be a normal good for another household • YED is an estimate calculated over separate time periods • Reserve 1 mark for a justified conclusion	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
2	With the help of a formula, explain the meaning of income elasticity of demand <u>and</u> consider the extent to which a rise in income will increase the consumption of <u>all</u> goods and services. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) An accurate definition of income elasticity of demand (1) and accurate formula (1). Understanding of the difference between normal and inferior goods (1). AO2 Analysis (max 3 marks) Normal goods have a positive value (1). Uses a diagram (upwards sloping income demand curve) or the formula to show why an increase in income increases the consumption of such goods (1) – this should be justified with an appropriate example (1). AO3 Evaluation (max 2 marks) One mark for a valid explanation that if the good is inferior good then the demand for the good will fall when incomes rise (1) as consumers adjust their spending habits and consumer more normal/luxury goods. The second mark is for a justified conclusion.	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
3	With the help of a formula, explain what is meant by the income elasticity of demand for a product and consider the extent to which demand for the product will always rise at the same rate as the income of its consumers. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) For an accurate formula of YED (1). For clear and accurate knowledge and explanation of what is meant by YED (1) including the difference between elastic and inelastic YED (1). AO2 Analysis (max 3 marks) For a clear understanding and explanation that this depends on the coefficient value of YED (1) and analysis of the different outcomes depending on whether the product is normal (positive coefficient) (1) or inferior (negative coefficient) (1). AO3 Evaluation (max 2 marks) For a considered evaluation of the possible outcomes (1) leading to a valid conclusion (1).	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

SECTION B

1.	A	8.	B
2.	D	9.	B
3.	B	10.	A
4.	C		
5.	D		
6.	B		
7.	B		